

Controlled growth of perovskite layers with volatile alkylammonium chlorides

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Jaewang Park^{1,5}, Jongbeom Kim^{1,5}, Hyun-Sung Yun¹, Min Jae Paik¹, Eunseo Noh¹,
Hyun Jung Mun², Min Gyu Kim^{3,✉}, Tae Joo Shin^{4,✉} & Sang Il Seok^{1,✉}

Controlling the crystallinity and surface morphology of perovskite layers by methods such as solvent engineering^{1,2} and methylammonium chloride addition^{3–7} is an effective strategy for achieving high-efficiency perovskite solar cells. In particular, it is essential to deposit α -formamidinium lead iodide (FAPbI₃) perovskite thin films with few defects due to their excellent crystallinity and large grain size. Here we report the controlled crystallization of perovskite thin films with the combination of alkylammonium chlorides (RACl) added to FAPbI₃. The δ -phase to α -phase transition of FAPbI₃ and the crystallization process and surface morphology of the perovskite thin films coated with RACl under various conditions were investigated through in situ grazing-incidence wide-angle X-ray diffraction and scanning electron microscopy. RACl added to the precursor solution was believed to be easily volatilized during coating and annealing owing to dissociation into RA⁰ and HCl with deprotonation of RA⁺ induced by RA⁺...H⁺-Cl⁻ binding to PbI₂ in FAPbI₃. Thus, the type and amount of RACl determined the δ -phase to α -phase transition rate, crystallinity, preferred orientation and surface morphology of the final α -FAPbI₃. The resulting perovskite thin layers facilitated the fabrication of perovskite solar cells with a power-conversion efficiency of 26.08% (certified 25.73%) under standard illumination.

So far, alkylammonium cations (RA⁺) with longer lengths or larger ionic radii than that of formamidinium (FA⁺) and methylammonium (MA⁺) have been mainly used in forming one-dimensional or two-dimensional (1D or 2D) lead halide perovskites^{8–10}, for coexisting with three-dimensional (3D) structures and defect passivation of grain boundaries and surfaces. For this purpose, iodide (I⁻) or bromide (Br⁻) is mainly used as a halide anion that binds to RA⁺. This is because the proton affinity of Cl⁻ ions is greater than that of I⁻ and Br⁻ ions, so that RA⁺Cl⁻ can easily dissociate into deprotonated RA⁰ and H⁺Cl⁻, which can then volatilize^{11–13}. On the basis of this volatilization, there have been many reports related to low-temperature formation, grain growth and preferential orientation of α -FAPbI₃ when perovskite thin films are deposited by adding shorter alkylammonium chloride (RACl), MACl, to the precursor solution^{3–7}. However, there are few reports of using longer RACl such as propylammonium chloride (PACl) and butylammonium chloride (BACl) as an agent for improving perovskite crystallization and surface morphology^{13–15}. The volatilization rate of propylamine (PA⁰) and butylamine (BA⁰) produced by deprotonation of PACl and BACl may be slower than that of MACl, because the boiling points of PA⁰ and BA⁰ are 47.8 °C and 78 °C, respectively, which are much higher than that of methylamine (MA⁰, -6.3 °C). Thus, the addition of these RACls with different physical properties is expected to have a great influence on the crystallization behaviour during the formation of perovskite thin films. Nonetheless, few studies have investigated whether the benefits of adding RACl are because of Cl⁻ ions, RA⁺ or the synergistic effects of both Cl⁻ ions and RA⁺.

This work reports the understanding of the role of each RA⁺ and Cl⁻ (specifically, RACl added on the basis of MACl) in the crystallization and surface morphology of the coatings of RACl-added FAPbI₃ precursor solutions and the perovskite solar cells (PSCs) fabricated from the resulting α -FAPbI₃ thin films.

Formation of α -FAPbI₃ thin films

Figure 1a shows the surface morphologies observed by scanning electron microscopy (SEM) for the thin films formed by dripping the antisolvent during the spin-coating process of the FAPbI₃ precursor solutions with 0, 5, 10 or 15 mol% PACl added to a 35-mol% MACl base. The thin films coated with the FAPbI₃ precursor containing only 35 mol% MACl without PACl showed distinct grain boundaries. Moreover, several pinholes mixed with the residual coating medium were observed. By contrast, the grain boundaries disappeared with increasing PACl and a uniform surface morphology without pinholes was obtained. Similarly, the surface morphology of the coated thin film with further addition of 10 mol% MACl or BACl to the precursor solution containing 35 mol% MACl also improved in terms of surface uniformity (Extended Data Fig. 1a), even though PACl was more uniform. These results indicate that the increase in RACl, which can bind to PbI₂ in FAPbI₃, decelerates the nucleation rate of the intermediate phase induced by dripping the antisolvent. Moreover, depending on the RA⁺ type, a uniform thin film with large grains is formed. For fabricating α -FAPbI₃-based

¹Department of Energy and Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea. ²Department of Materials Science and Engineering, Chonnam National University, Gwangju, Korea. ³Beamline Research Division, Pohang Accelerator Laboratory (PAL), Pohang University of Science and Technology (POSTECH), Pohang, Republic of Korea.

⁴Graduate School of Semiconductor Materials and Devices Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea. ⁵These authors contributed equally:

Jaewang Park, Jongbeom Kim. ✉e-mail: mgkim@postech.ac.kr; tjshin@unist.ac.kr; seoksi@unist.ac.kr

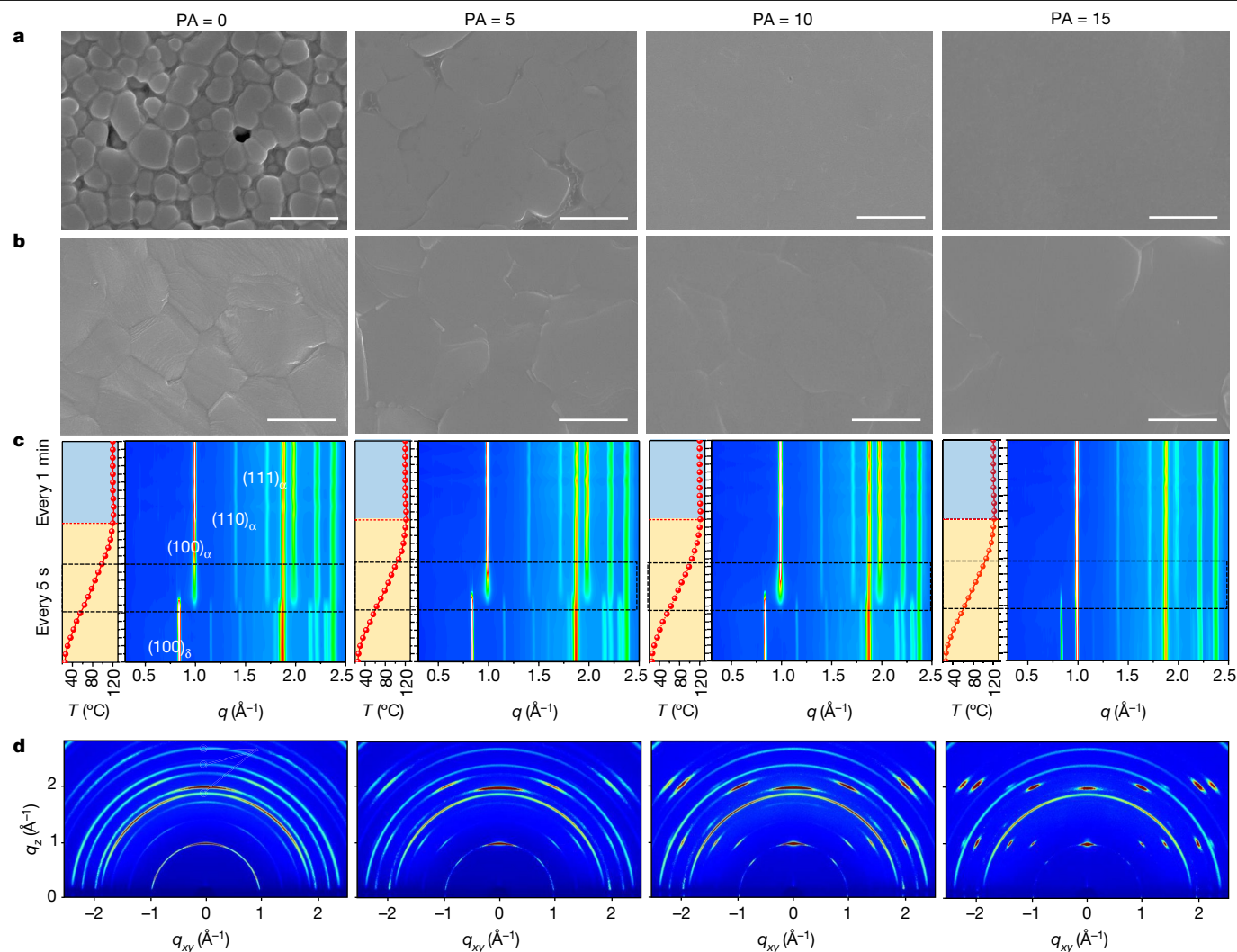


Fig. 1 | Changes in the surface morphology and crystal phase of perovskite thin films with volatile RACl. **a**, Surface morphologies of the thin films spin-coated with 0–15 mol% PACl added to the FAPbI₃ precursor containing 35 mol% MACl. **b**, Surface morphologies after annealing them at 120 °C for 40 min.

c, The structural evolution monitored by azimuthally integrated in situ GI-WAXD during heating from RT to 120 °C using the as-deposited thin films. **d**, 2D GI-WAXD of the perovskite thin films annealed at 120 °C for 30 min and cooled to RT. Scale bars, 1 μm .

high-efficiency PSCs, it is important to maintain the uniformity of the intermediate thin film, even after high-temperature heat treatment. Therefore, the as-deposited thin films were annealed at 120 °C for 40 min and then compared with the surface morphology. As shown in Fig. 1b, after annealing, all thin films were formed into very dense and pinhole-free uniform thin films, with or without non-uniformities, such as large pores observed in the as-deposited thin films. The surface morphology of the thin film coated with the precursor, to which only MACl was added, was identical to those reported in the previous study¹⁶. In particular, thin films coated with precursors containing 10 mol% and 15 mol% PACl exhibited an extremely flat surface morphology even after annealing compared with that of the precursors containing 10 mol% MACl or BACl (Extended Data Fig. 1b). We further compared the surface morphology and roughness of perovskite thin films deposited with precursors containing an extra 10 mol% of RACl using atomic force microscopy (Supplementary Fig. 1). It was observed that the addition of 10 mol% of RACl to the FAPbI₃ precursor containing 35 mol% of MACl reduced the roughness of the thin film surface but the smoothest surface was obtained with PACl. Thus, PACl added to 35 mol% MACl plays a decisive role in improving the surface morphology, leading to improved efficiency. Figure 1c shows the structural evolution monitored by in situ grazing-incidence wide-angle X-ray diffraction

(GI-WAXD) during heating from room temperature (RT) to 120 °C using the as-deposited thin films. The initial δ -FAPbI₃ thin film was converted to α -FAPbI₃ in a heating regime of approximately 60–80 °C. Notably, the as-deposited thin film coated with the precursor to which 15 mol% PACl was added contained α -FAPbI₃ and δ -FAPbI₃ at RT. Generally, α -FAPbI₃ requires a heating temperature above 150 °C to achieve a corner-shared PbI₆ octahedral structure^{17–19}. Presumably, α -FAPbI₃ at low temperatures of 60–80 °C and even RT can be attributed to the MACl-induced kinetic effect, such as the PbI₆ octahedral rotation^{20,21} and synergistic effect of PACl coexisting with MACl, as will be discussed later. It is known that adding 100 mol% MACl, approximately three times the amount used in this study, to the FAPbI₃ precursor produces an α phase with a crystal orientation along the (100)_a plane at RT immediately after the evaporation of the coating solvent²². This can be related to the formation of unstable 2D MAPbI₃Cl (ref. 22) and coordination of Cl[−] ions at the octahedra terminal corner that inhibits the face-shared octahedron, which is the δ -FAPbI₃ structure. Zhao and colleagues²³ reported that HCl promoted the δ -phase to α -phase transition at RT but the transition to the α phase at RT became similar to that with MACl only when the evaporation rate was controlled after adding 100 mol%, which was the same amount as MACl. Therefore, it is reasonable to conclude that the content of Cl[−] ions added to the

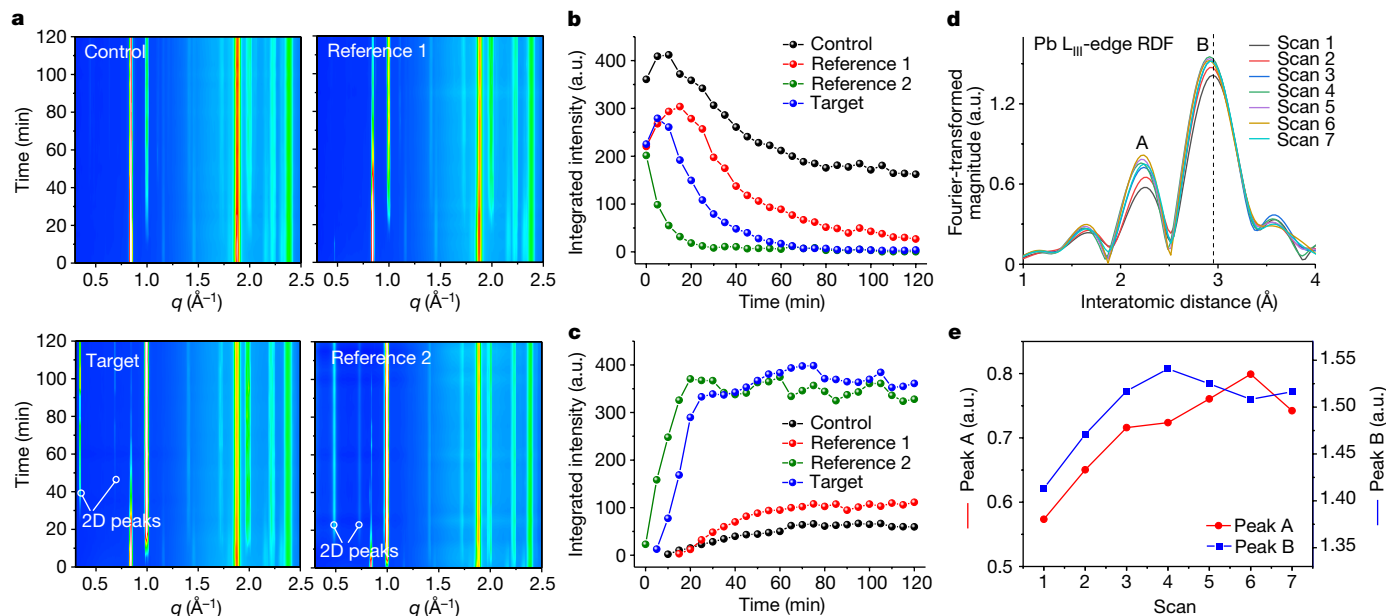


Fig. 2 | Structural evolution by in situ GI-WAXD of thin films coated with volatile RACl. **a**, The structural evolution of the as-deposited thin films, which were coated using the FAPbI₃ precursor solutions (control, reference 1, reference 2 and target), by azimuthally integrated in situ 2D GI-WAXD during ageing at RT for 120 min. **b**, The integrated intensity of δ -FAPbI₃ extracted from 2D GI-WAXD over time. **c**, The integrated intensity of α -FAPbI₃ extracted from

2D GI-WAXD over time. **d**, Pb L₃-edge EXAFS spectra of the as-deposited thin films, which were coated using the FAPbI₃ precursor solution containing MACl and a further 10 mol% of PACl, during ageing at RT for 120 min. The Fourier transform of the *k*-space EXAFS spectra was plotted. **e**, The plot of Fourier-transformed peak intensities of Pb-Cl (A) and Pb-I (B) corresponding to the structural development of the EXAFS spectra (in **d**). a.u., arbitrary units.

perovskite precursor and the control of the release rate of Cl[−] ions by RA⁺ to which Cl[−] ions are bound are important factors determining the δ -phase to α -phase transition. Notably, in the thin film coated with the precursor to which 5 mol% and 10 mol% PACl were added, the width of the (100)_α peak ($q \approx 10 \text{ nm}^{-1}$) became much broader at 60–80 °C, at which the transition to the α phase started, and disappeared as the temperature increased. The broadening of the (100)_α peak is more pronounced in a thin film containing 10 mol% PACl than that containing 5 mol% and was absent in the film with 15 mol% of PACl, which showed a large amount of α -FAPbI₃ even before annealing (Fig 1c and Extended Data Fig. 2). The (100)_α peak width variation at the δ -phase to α -phase transition temperature is the greatest in thin films coated with precursors doped with 10 mol% PACl, which may provide further kinetics for crystallization into high-quality α -FAPbI₃. Figure 1d compares the 2D GI-WAXD of the perovskite thin films obtained after annealing the as-deposited thin film at 120 °C for 30 min. A small amount of the intermediate phase containing MACl and PACl and the solvent and δ phase disappeared after heat treatment, transforming into a nearly pure α -FAPbI₃ perovskite structure. This observation is also supported by the time-of-flight secondary ion mass spectrometry (ToF-SIMS) results, which showed that almost no Cl[−] ions remained in FAPbI₃ in all thin films (Supplementary Fig. 2). As reported for MACl addition^{7,14,16}, few residual Cl[−] ions were found in PACl and BACl, indicating that almost all of the added RACl evaporated during the coating and annealing processes. The reason for the ready disappearance of RACls added to the precursor solution, such as MACl and PACl, whereas only FAPbI₃ remains, will be discussed later. In thin films obtained from precursors containing PACl, a preferred orientation along the (100)_α direction and crystallinity increased as PACl increased up to 15 mol%, with a slight difference between 10 mol% and 15 mol%. This result indicated that PACl led to sharp Bragg diffraction spots of the (100)_α plane along the Debye–Scherrer ring at $q \approx 10 \text{ nm}^{-1}$, indicating improved crystallinity and grain orientation in the out-of-plane direction perpendicular to the substrate. The influence of these crystallinities, crystal orientation and surface morphology controls on the device performance was

investigated according to the amount of PACl added to the precursor solution. As shown in Supplementary Fig. 3, the average efficiency was the highest in the devices fabricated with a precursor solution wherein 10 mol% PACl was added to 35 mol% MACl. The efficiency improvement with PACl indicates that the effect of PA⁺ cations, rather than that of Cl[−] ions, is dominant because the PSCs fabricated with the precursor to which more MACl was added resulted in lower performance. This implies that more controlled crystallization occurs in the perovskite thin film, in which 10 mol% of PACl is added to the precursor.

δ -phase to α -phase transition of FAPbI₃

To gain further insight into the PA⁺ cations for the δ -phase to α -phase transition at low temperatures and highly flat surface morphology of the FAPbI₃ perovskite thin films, we first monitored the surface and structural evolution of the as-deposited thin films at RT. Their behaviours were compared with those of thin films coated with precursors to which 10 mol% MACl and BACl were added because the highest efficiency was obtained when 10 mol% PACl was added to a FAPbI₃ precursor solution containing 35 mol% MACl. Figure 2a shows azimuthally integrated in situ 2D GI-WAXD measured during ageing of as-deposited film at RT for 0–120 min. Hereafter, 35 mol% of MACl in the FAPbI₃ precursor solution is referred to as ‘control’, further addition of 10 mol% of MACl and BACl as ‘reference 1’ and ‘reference 2’, respectively, and further addition of 10 mol% of PACl as ‘target’. In the film coated with the control, the δ phase was almost 100% and the α phase appeared after several tens of minutes and increased substantially during ageing for 120 min. However, even after 120 min, the δ phase was dominant. The as-deposited thin film coated with reference 1 exhibited a greater δ -phase to α -phase transition tendency but the δ phase remained after ageing for 120 min. By contrast, the thin film coated using the target and reference 2 rapidly underwent a δ -phase to α -phase transition with the generation of some amount of 2D phase as the ageing time increased and almost attained the α -phase after 120 min. Because the in situ-generated 2D perovskites disappeared during annealing (Fig. 1c), both

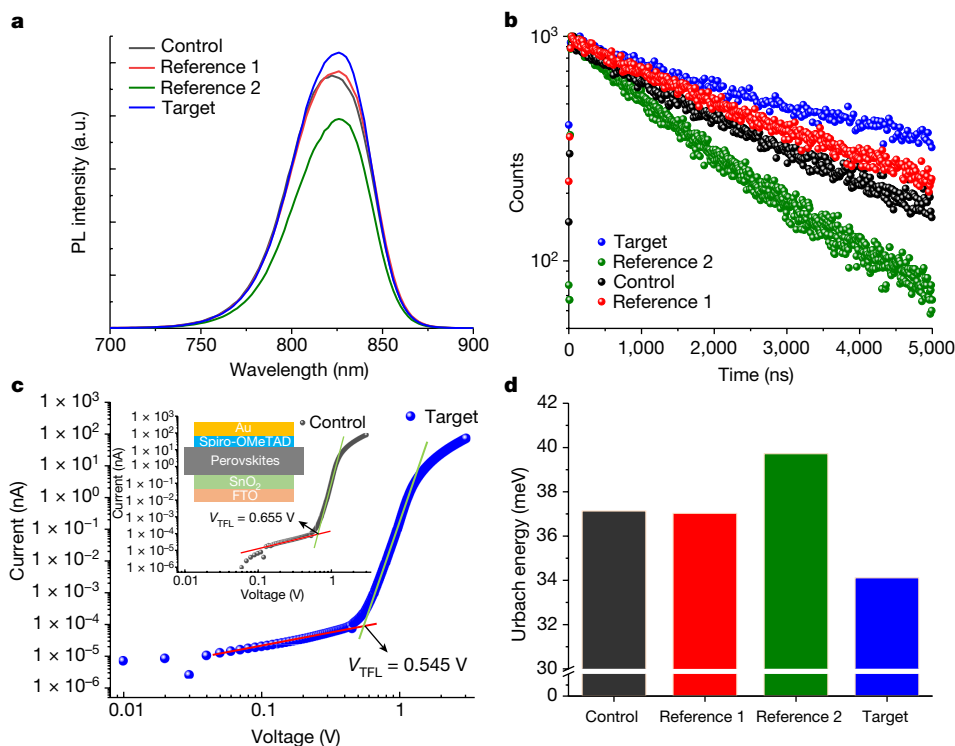


Fig. 3 | Characteristics of perovskite layers deposited with volatile RACl.

a,b, Steady-state PL (**a**) and time-resolved PL (**b**) spectra of films deposited on a glass substrate. **c**, Space-charge-limited current analysis for the target and

control (inset). V_{TFL} , trap-filled limit voltage. The inset schematic shows the structure of the device used for the measurements. **d**, Urbach energy calculated in perovskites consisting of FTO/SnO₂/perovskites. a.u., arbitrary units.

PA⁺ and BA⁺ cations, like MA⁺ cations, presumably escaped from the thin film. The time-dependent change in the α and δ phases shown in the GI-WAXD of Fig. 2a is quantitatively illustrated in Fig. 2b,c. The integral of the δ phase for each ageing time initially increased and then rapidly decreased, with the change being more pronounced over time (Fig. 2b). Concurrently, the α phase rapidly increased from the beginning and was saturated from 30 min to 1 h. In the coating film of the precursor to which BACl was added, the δ phase sharply decreased, whereas the α phase rapidly increased from the beginning of ageing (Fig. 2c); thus, the two changes were well matched. On this addition of BACl, the faster δ -phase to α -phase transition is believed to be associated with a slower volatilization rate of deprotonated BA⁰ and stronger base–acid interaction with PbI₂ of FAPbI₃, because the temperature of such a phase transition is also the lowest in BACl (Extended Data Fig. 3). However, such a δ -phase to α -phase transition at RT did not occur unless an antisolvent was dropped during the spin-coating process of the precursor solution on the substrate, and the phase transition occurred only when the temperature was raised (Extended Data Fig. 4). As shown by comparing the GI-WAXD (Fig. 1c) for the target thin film as deposited with and without dripping the antisolvent, a marked difference was observed in terms of crystallinity and orientation and unknown intermediates. Furthermore, in situ GI-WAXD measurements during annealing to 120 °C with the previous two coating films demonstrated that the α phase in the thin film coated without dripping the antisolvent appeared at a relatively high temperature, greater than 70 °C (Extended Data Fig. 4). The differences could be related to the washout of almost all dimethyl sulfoxide (DMSO) or N,N-dimethylformamide (DMF) bound to FAPbI₃ when the antisolvent was dropped during spin coating using RACl-added precursors. This observation can be attributed to the fact that, in RACl and DMSO, which compete with PbI₂ as the Lewis base, DMSO is soluble in the antisolvent, whereas RACl remains insoluble; thus, the RACl may dominate the crystallization of the perovskite. Particularly, unlike the intermediate of FAPbI₃-DMSO, RACl coordinated

to FAPbI₃ is considered present on the surface of crystalline FAPbI₃ precipitated by the dripping of an antisolvent. On this basis, the behaviour seen in Fig. 2a can be explained by the presence of RACl on the surface of the δ -phase crystal, which reduces the surface energy, converting to the corner-shared structure from the edge-shared phase by rotating the PbI₆ octahedron^{20,21,24}, and can further orient the crystal in the (100)_α direction by orientation attachment^{25,26}. Numerous studies have reported that several substances, such as Cl[−] and SCN[−] bound to the surface of δ -FAPbI₃, lower the transition barrier, thereby facilitating the transition to the α phase at low temperatures^{20,23,27}. DMSO also facilitates the δ -phase to α -phase transition when adsorbed on the surface of the δ -FAPbI₃ crystals^{28,29}. In this context, the in situ GI-WAXD measurements demonstrate that δ -FAPbI₃ transforms to α -FAPbI₃ at 60–80 °C during slow heating. Thus, we conclude that the phase transition is generally thermodynamically favourable above 60 °C because of the lowering of the activation barrier in the presence of RACl. At RT, the barrier kinetically suppresses the return to the δ phase²⁰. The visual colour change also confirms the δ -phase to α -phase transition with ageing of the as-deposited thin film. In the digital photograph of the change with time of the same thin film as shown in Fig. 2a, the black areas are darkened owing to the formation of α -FAPbI₃ according to the type of RACl (Extended Data Fig. 5). Notably, the thin film coated without the antisolvent did not turn black, regardless of RACl type (Supplementary Fig. 4). The δ -phase to α -phase transition, occurring thermodynamically at a high temperature of 150 °C or even at RT, is attributed to the complex action of the metastable intermediate depending on the amount of Cl[−] ions and the type of RA⁺ with the removal of the residual solvents. There have been many reports on the δ -phase to α -phase transition and preferred orientation at low temperatures when adding Cl[−] anions, such as HCl and MACl (refs. ^{3,23,30,31}). Furthermore, the 2D phases resulting from RA⁺ lower the δ -phase to α -phase transition temperature and enhance the α -phase stability³². RACl added to the FAPbI₃ precursor solution substantially lowered the

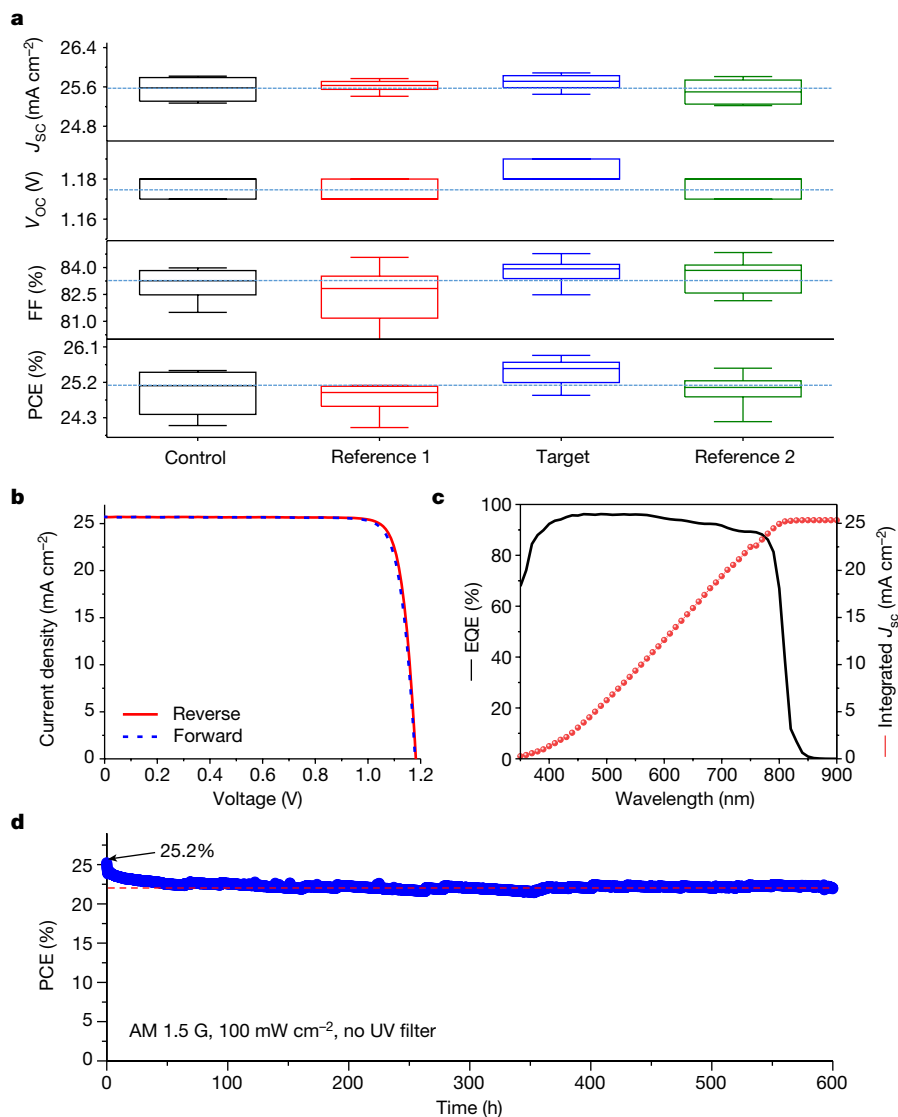


Fig. 4 | Performance and stability of PSCs, measured by applying an antireflection film to the surface. a, Statistical distribution of photovoltaic parameters for each of 24 PSCs fabricated with control, reference 1, reference 2 and target thin films. The box plots show the average value (centre line in box), the maximum and minimum values (whiskers), and the lower quartile (25%) and upper quartile (75%) (bottom and top of box). The horizontal dashed blue

lines allow for comparison against the control. **b**, $J-V$ curves of the best-performing PSC fabricated with the target, which was measured in the reverse and forward modes under AM1.5 G, including UV. **c**, The EQE curve of the corresponding PSC. **d**, Maximum power point tracking for a PSC fabricated with the target, which was measured under full solar illumination (AM1.5 G, 100 mW cm⁻² under ambient conditions) without a UV filter.

transition temperature, which even disappeared during ageing at RT (Supplementary Fig. 5). This result indicates that the δ -phase to α -phase transition of the coated thin film is kinetically affected by the amount of RACl and the type of RA⁺. The transformation of α -FAPbI₃ perovskite from the crystalline precursor film was also investigated in the formation chemistry by measuring the extended X-ray absorption fine structure (EXAFS) on the Pb L_{III} edge³³. Consistent with the behaviour in GI-WAXD (Fig. 2a), the interatomic distance between Pb–I and its peak intensity, indicating the δ -phase to α -phase transition (Extended Data Fig. 6), was changed (Fig. 2d). Furthermore, the number of I⁻ and Cl⁻ ions coordinating with Pb²⁺ cations increased in response to this phase transition and then began decreasing (Fig. 2e). This result implies that the coordination of RACl to PbI₂ initially increases, leading to the formation of PbI₆ octahedrons, and then slowly disappears. RA⁺ (soft acid) acts more like a soft base (I⁻) to form stable products, whereas RA⁺ (soft acid) and Cl⁻ (hard base) form unstable products. Furthermore, the primary ammonium (RA⁺) ions, although their pK_a slightly depends on the length of the alkyl group, are weakly acidic and deprotonate more

readily than FA⁺ (ref. 34). Thus, the RACl coordinated to PbI₂ (or FAPbI₃) was expected to be converted to highly volatile species (for example, MA⁰, PA⁰ and HCl) through processes such as deprotonation of RA⁺ (refs. 11,12,34) while binding with PbI₂ as RA⁺ cations and Cl⁻ anions (RA⁺...H⁺·Cl⁻–PbI₂ + FA⁺ → RA⁰↑ + HPbI₂Cl + FA⁺ → HCl↑ + FAPbI₃). Indeed, RACl itself decomposes and evaporates only above 180 °C (ref. 35). The RA⁰ generated by RA⁺ deprotonation may exist in a liquid phase before volatilization during annealing, depending on its boiling point. Considering this, we surmise that liquid RA⁰ can influence the surface flattening of FAPbI₃, depending on the RA⁰ type. As expected, the BACl-added precursor resulted in a better surface morphology flatness than that obtained using MACl (Extended Data Fig. 1b); however, the thin film coated with PACl-added precursor was the flattest.

Characteristics of α -FAPbI₃ thin films

Figure 3a shows the photoluminescence (PL) spectra of perovskite layers annealed at 120 °C for 40 min after spin-coating the precursor

solutions with different types and amounts of volatile RACl on a glass substrate. The control, reference 1 and reference 2 have similar PL intensities but that of the target is slightly high and implies decreasing nonradioactive recombination. For a more quantitative comparison of charge recombination, time-resolved PL was measured using time-correlated single-photon counting under low-intensity pulsed excitation, in which a mono-exponential decay through nonradiative-trap-induced recombination can be observed³⁶. The corresponding mono-exponential lifetimes (Supplementary Table 1) were obtained from the PL decay curves in Fig. 3b. The carrier lifetime of the target film ($\tau = 2,929$ ns) was longer than that of the control, reference 1 and reference 2 films ($\tau = 2,187, 2,582$ and $1,503$ ns, respectively). The highest PL intensity and the longest lifetime in the target film indicated that the nonradiative recombination was the lowest, suggesting the formation of a thin film with the lowest defect concentration by adding 10 mol% PACl. We further measured the dark current–voltage of the full devices by means of the space-charge-limited current method^{37,38} (Fig. 3c and Extended Data Fig. 7). From the calculated trap densities (Supplementary Table 2), the trap densities are calculated to be 1.13×10^{16} , 1.05×10^{16} , 1.18×10^{16} and 9.37×10^{15} cm⁻³ for the control, reference 1, reference 2 and target films, respectively. Obviously, the target showed the lowest trap concentration, indicating that the defects in the perovskite film coated with the 10-mol%-PACl-added precursor were reduced. Although target and reference 2 are almost similar in terms of surface morphology and the δ -phase to α -phase transition of the perovskite thin film, the lowest trap density in target is believed to be because of higher crystallinity than reference 2. This may be related to the relatively too fast rate of crystallization in the formation of the perovskite thin film of reference 2 (Extended Data Fig. 8). Urbach energies (E_u) were also compared because the low defect concentration of the perovskite thin film responsible for the high performance was because of the superiority of the structural quality during the crystal formation of perovskites. E_u for the perovskite films was calculated from the ultraviolet (UV)–visible absorption spectra using the following equation: $\alpha = \alpha_0 \exp(h\nu/E_u)$, in which α is the absorption coefficient and $h\nu$ is the photon energy. E_u in the target represented the lowest value, indicating the highest structural quality (Fig. 3d and Supplementary Fig. 6).

Photovoltaic performance

The consistent results that the target thin film is of high quality encourage further investigation into the benefits of thin films in photovoltaic devices. Thus, 24 PSCs under identical conditions using different RACls were fabricated to confirm their impact on performance in each category. The statistical distribution of current density (J_{sc}), open-circuit voltage (V_{oc}) and fill factor (FF) determining the power-conversion efficiency (PCE) are shown in Fig. 4a. As expected from the defect concentration and carrier lifetime analysis of the perovskite thin films formed under each condition, the average PCE of target was the highest and increased compared with that of reference 2. By contrast, the average PCE of reference 1 was lower than that of control. Excessive addition of MACl may be related to the incorporation of small amounts of MA⁺ ions into FAPbI₃ (ref. 7), which affects the bandgap (Supplementary Fig. 7) and external strain³⁹, further affecting the reduction of J_{sc} and FF and, eventually, PCE. The J – V characteristics of the best-performing target device in reverse and forward bias sweeps with antireflective coatings on the device surface are illustrated in Fig. 4b. The J_{sc} , V_{oc} and FF values calculated from the J – V curve at a reverse scan mode were 25.69 mA cm⁻², 1.178 V and 86.15%, respectively; these factors correspond to the PCE of 26.08% under standard air mass 1.5 global (AM 1.5 G) conditions, with J_{sc} , V_{oc} , FF and PCE values of 25.64 mA cm⁻², 1.178 V, 85.06% and 25.7%, respectively, at the forward bias. The PCE of the device was further verified by an accredited laboratory (Newport, USA) using

the quasi-steady-state method (Supplementary Fig. 8). The resulting stabilized efficiency measured was 25.73%, with J_{sc} , V_{oc} and FF values of 25.80 mA cm⁻², 1.179 V and 84.60%, respectively, which is the highest efficiency ever reported. Figure 4c shows the external quantum efficiency (EQE) for the same device as that of the J – V curve. The J_{sc} obtained by integrating the EQE over the wavelength is 25.3 mA cm⁻², which is in good agreement with that in the J – V curve measured using the solar simulator. When tested with maximum power point tracking under ambient and full sunlight illumination without a UV cutoff filter, the encapsulated target device retained approximately 88% (25.2%) of its initial efficiency after 600 h (Fig. 4d).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-05825-y>.

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Methods

Materials

The following materials were used in the experiments: formamidinium iodide (FAI; 99.99%, GreatCell Solar), lead(II) iodide (PbI_2 ; 99.99%, TCI), methylenediamine dihydrochloride (MDACl_2 ; 98%, Sigma-Aldrich), methylammonium chloride (MACl; Merck), propylamine hydrochloride (PACl; 99%, Sigma-Aldrich), butylamine hydrochloride (BACl; 98%, TCI), 4-methoxy-phenethylammonium iodide (MeO-PEAI; 99%, GreatCell Solar), tin(II) chloride dihydrate ($\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$; 98%, Acros), Urea (Alfa Aesar), deionized water (H_2O ; Alfa Aesar), potassium chloride (KCl; 99.999%, Sigma-Aldrich), hydrochloric acid (HCl; 37 wt% in H_2O , Sigma-Aldrich), thioglycolic acid ($\text{C}_2\text{H}_4\text{O}_2\text{S}$; 99%, Sigma-Aldrich), N,N-dimethylformamide (DMF; anhydrous, 99.8%, Sigma-Aldrich), dimethyl sulfoxide (DMSO; anhydrous, 99.9%, Sigma-Aldrich), 2-propanol (anhydrous, 99.5%, Sigma-Aldrich), 2-methoxyethanol (2-ME; anhydrous, 99.8%, Sigma-Aldrich), chlorobenzene (anhydrous, 99.8%, Sigma-Aldrich), acetonitrile (ACN; anhydrous, 99.8%, Sigma-Aldrich), Spiro-OMeTAD (Lumtec), 4-*tert*-butylpyridine (t-BP; 98%, Sigma-Aldrich), bis(trifluoromethane)sulfonimide lithium salt (Li-TFSI; 99.95%, Sigma-Aldrich), FK209 Co(III) TFSI salt (Lumtec) and ethyl ether (99.0%, SAMCHUN).

Formamidinium lead triiodide (FAPbI_3) black powder was synthesized by mixing 3.36 g FAI with 9 g PbI_2 (1:1 molar ratio) in 11 ml of 2-ME in a 70-ml vial by stirring. The mixed solution was heated to 120 °C in an oil bath and stirred for 45 min. Subsequently, the precipitated FAPbI_3 was filtered using filter paper (Whatman) without cooling to RT. The filtered FAPbI_3 was baked on a hot plate at 150 °C (in air) for 30 min and then left overnight in a vacuum chamber at RT.

Device fabrication

PSCs were composed of several layers of glass/fluorine-doped tin oxide (FTO)/ SnO_2 /perovskite/2,2',7,7'-tetrakis[*N,N*-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (Spiro-OMeTAD)/Au. FTO glass (Asahi VU glass, 12–13 $\Omega \text{ cm}^{-2}$) substrates were sequentially sonicated in acetone, detergent and ethanol for 20 min each. The FTO glasses were dipped in a solution prepared by mixing $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ (108 mg), urea (500 mg), thioglycolic acid (10 μl) and HCl (500 μl) in 40 ml of deionized water for 6 h. The substrates were then annealed at 150 °C for 1 h, followed by spin-coating with KCl solution (40 mM) in deionized water and annealing at 150 °C for 15 min. The perovskite precursor solutions were prepared by dissolving 1.4 M FAPbI_3 powder with MDACl_2 (3.8 mol%) and MACl (35 mol%) in a mixed solvent of DMF:DMSO = 8:1 (control); 10 mol% of different alkylammonium chlorides were added for the reference 1 (MACl 10 mol%), reference 2 (BACl 10 mol%) and target (PACl 10 mol%) solutions. The perovskite solutions were then spin-coated onto the substrate at 1,000 rpm for 10 s and 5,000 rpm for 15 s; 1 ml of ethyl ether was dripped onto the substrate during spinning. Before all perovskite layers were heat-treated at 120 °C for 1 h, the as-deposited layers were aged for a period of time at RT, depending on the added RACl.

Characterization

The surface morphology of the perovskite layers was observed by SEM (Cold FE-SEM SU-8220, Hitachi High-Technologies) with a beam power of 5.0 kV. Steady-state PL and time-resolved PL spectra were measured using a commercial time-correlated single-photon-counting setup (FluoTime 300, PicoQuant GmbH) equipped with a PMA-C-192-M detector and high-resolution excitation monochromators. The J - V characteristics of the devices were measured using a Keithley 2420 source meter under illumination by a solar simulator (Newport, Oriel Sol3A class AAA) with an AM 1.5 G filter and 100 mW cm^{-2} irradiation intensity. The active area was determined by placing a metal mask in front of the solar cell to avoid the overestimation of the photocurrent density. A spectral mismatch factor of 1.05 was used for all J - V

measurements. For the measurement of high-efficiency devices, an antireflective film (NanoEcoWay Co., Ltd.) was applied to the surface. The EQE was measured using an internal quantum efficiency system (Oriel, IQE 200B) under irradiation by a 100-W xenon lamp. The film morphology and thickness were measured using SEM (FE-SEM, Su-7000).

GI-WAXD measurements

GI-WAXD measurements were performed at the PLS-II 6D UNIST-PAL beamline at the Pohang Accelerator Laboratory (PAL), Korea. The X-rays from the bending magnet were monochromatized to 18.986 keV ($\lambda = 0.6530 \text{ \AA}$) using a double-crystal monochromator focused horizontally and vertically ($90 \text{ (V)} \times 120 \text{ (H)} \mu\text{m}^2$ full width at half maximum) at the sample position using a sagittal Si(111) crystal and toroidal mirror, respectively. The GI-WAXD sample chamber was equipped with a five-axis motorized stage for fine sample alignment. The 2D GI-WAXD pattern was recorded using a 2D charge-coupled device detector (MX225-HS, Rayonix, LLC, USA). The diffraction angles were calibrated using NIST SRM660b (LaB_6) with a sample-to-detector distance of 240.98 mm. An incidence angle of 0.80° was used for the in situ GI-WAXD experiments to obtain crystallographic information about the overall thickness of the perovskite films and to ensure that all incoming X-rays were irradiated onto the small-length perovskite film sample, rather than only passing or being blocked by the substrate or sample stage. The in situ GI-WAXD pattern was obtained every 5 s for the initial 1.5 min and every 1 min between 1.5 and 30 min, in which the X-ray exposure time was 3.5 s each. The FAPbI_3 film temperature was increased from RT to 120 °C in 1.5 min and maintained at 120 °C for 30 min using a fine-tuned temperature controller (Eurotherm 3504 temperature controller, Eurotherm, USA).

EXAFS measurements

The specimen for EXAFS analysis was prepared by coating FAPbI_3 precursor solution containing MACl 35 mol% and PACl 10 mol%. The as-deposited layer was subjected to EXAFS for a total of six times, 2 min every 20 min. (The total X-ray exposure time was 12 min.) Pb L_{III} -edge X-ray absorption spectra, EXAFS, were collected on the BL10C beam line (WEXAFS) at the Pohang Light Source (PLS-II) using top-up mode operation under a ring current of 350 mA at 3.0 GeV. From the high-intensity X-ray photons of the multipole wiggler source, monochromatic X-ray beams could be obtained using a liquid-nitrogen-cooled double-crystal monochromator (Bruker ASC) with a Si(111) crystal pair. Under grazing incident angle of 3°, all X-ray absorption spectra were recorded in fluorescence mode using a N_2/Ar mixed-gas-filled ionization chamber (IC Spec, FMB Oxford) and passivated implanted planar silicon (PIPS, Canberra Co.) for the incident and fluorescent X-ray photons, respectively. Higher-order harmonic contaminations were eliminated by detuning to reduce the incident X-ray intensity by about 30%. Energy calibration was simultaneously performed for each measurement with a reference Pb foil in front of the third ion chamber using partially bypassed X-rays. Using the UWXAFS package, Fourier-transformed radial distribution functions (RDFs) of k^3 -weighted Pb L_{III} -edge EXAFS spectra, $k^3\chi(k)$ were obtained in the k range between 3.0 and 13.0 $\text{\AA}^{-1}$ through a standard XAFS data-analysis process. Ideal RDFs for model δ -phase and α -phase FAPbI_3 could be obtained from ab initio calculation using FEFF9 code.

ToF-SIMS measurements

ToF-SIMS profiling measured the depth distributions of the Cl^- ions with perovskite/ SnO_2 on the FTO substrate. The samples were analysed using a ToF-SIMS V instrument (IONTOF) with a Bi^+ primary beam (25 keV and 1 pA) and Cs^+ sputter beam (0.25 keV and 12 nA). The sputter size was $200 \times 200 \mu\text{m}$; the analysis areas were $100 \mu\text{m} \times 100 \mu\text{m}$ (for depth profiling) and $500 \mu\text{m} \times 500 \mu\text{m}$ (for surface analysis); the area dosage of ions was $\text{PIDD } 2.48\text{e}^+$ (ref. ¹⁴).

Article

Device stability testing

The stability test at the continuous maximum power point tracking device operation under 1 sun illumination at AM1.5G (xenon lamp, 450 W) for the encapsulated devices was performed under an inert atmosphere (N_2) by fixing the voltage at V_{MPP} and then tracking the current output.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request.

Code availability

The code used for this study is available from the corresponding authors on reasonable request.

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Author contributions J.P., J.K. and S.I.S. conceived this work and designed the experiment. H.-S.Y. performed a preliminary experiment. J.P. and J.K. fabricated the PSCs with various electrodes and characterized the perovskite films. M.J.P. and E.N. prepared the electrodes and samples for analysis. H.J.M. and T.J.S. conducted and interpreted the GI-WAXD measurements. M.G.K. measured and interpreted the EXAFS measurements. S.I.S. wrote the draft of the manuscript and all authors contributed feedback and comments for revising the manuscript. S.I.S. directed and supervised the study.

Competing interests The authors declare no competing interests.

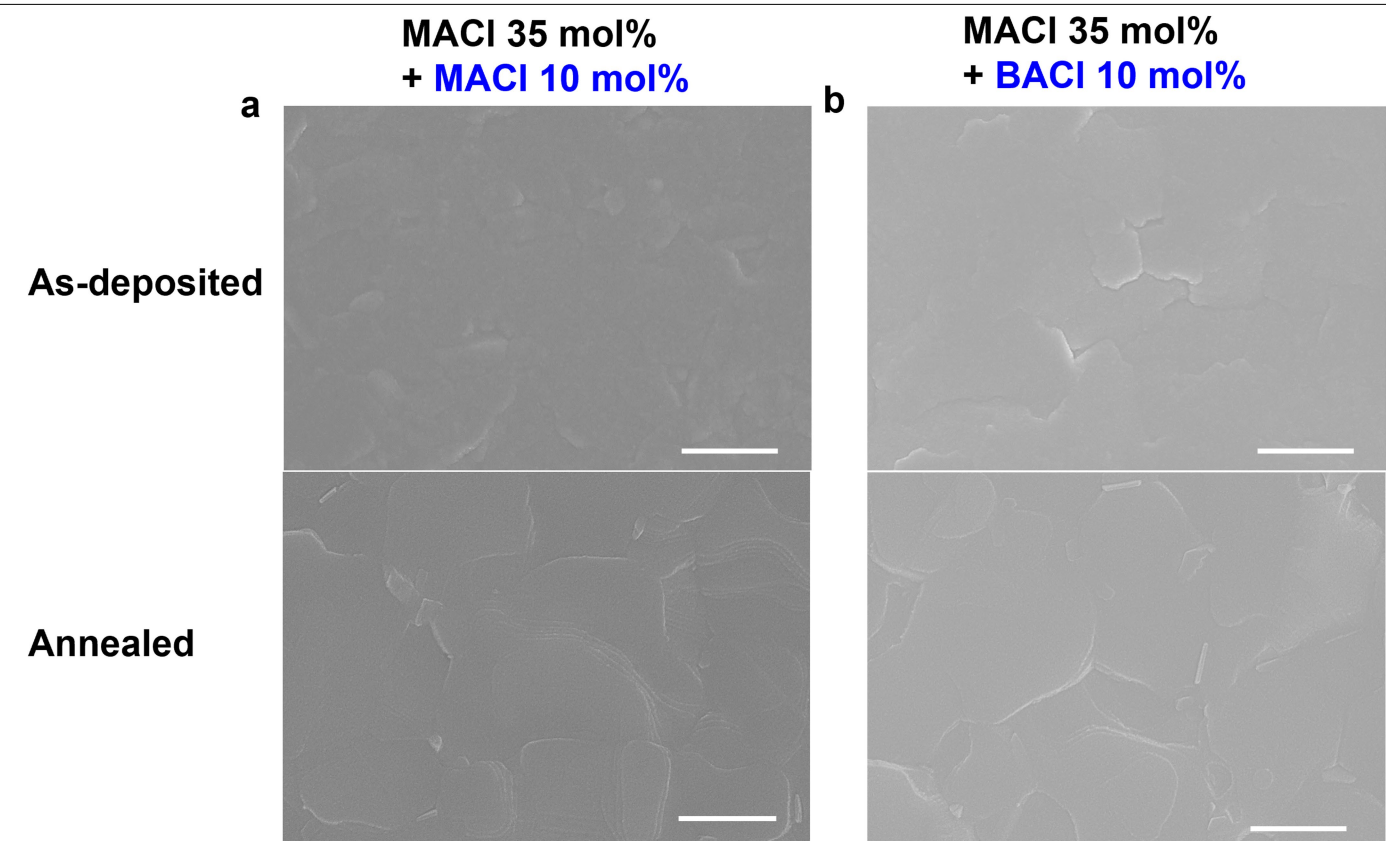
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-023-05825-y>.

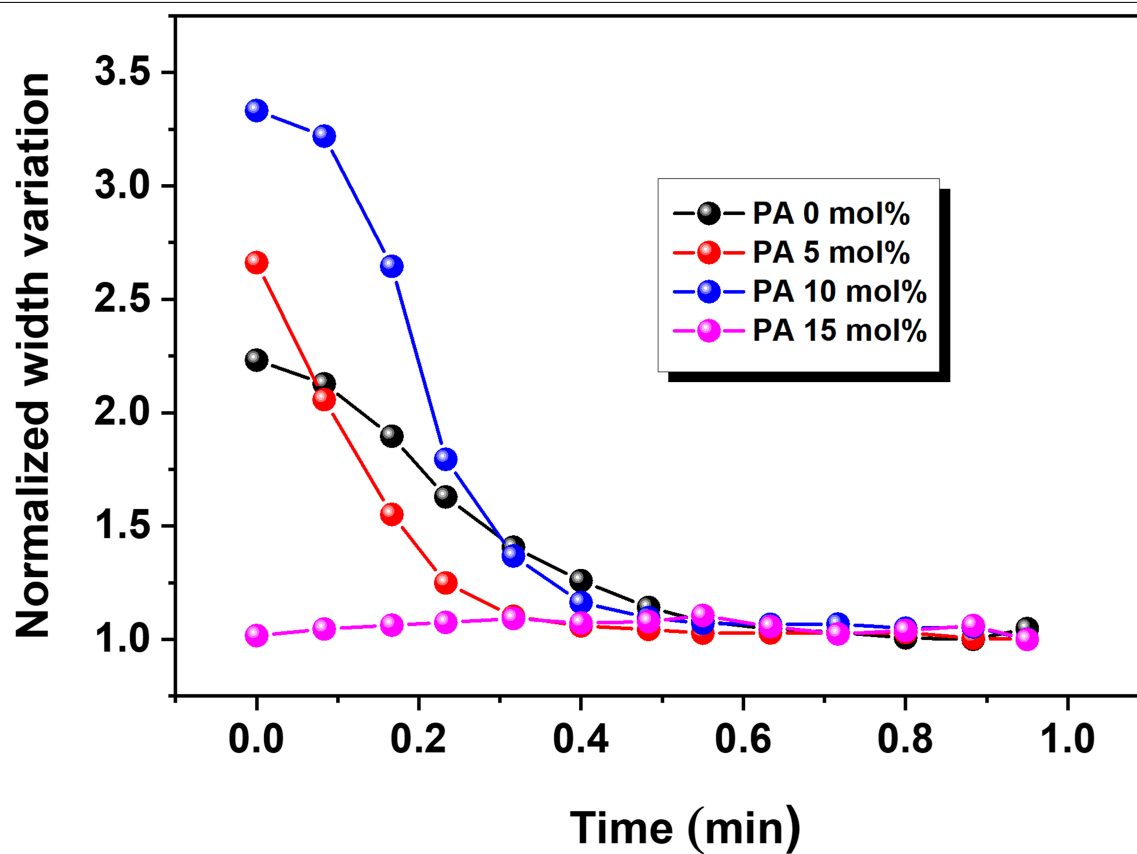
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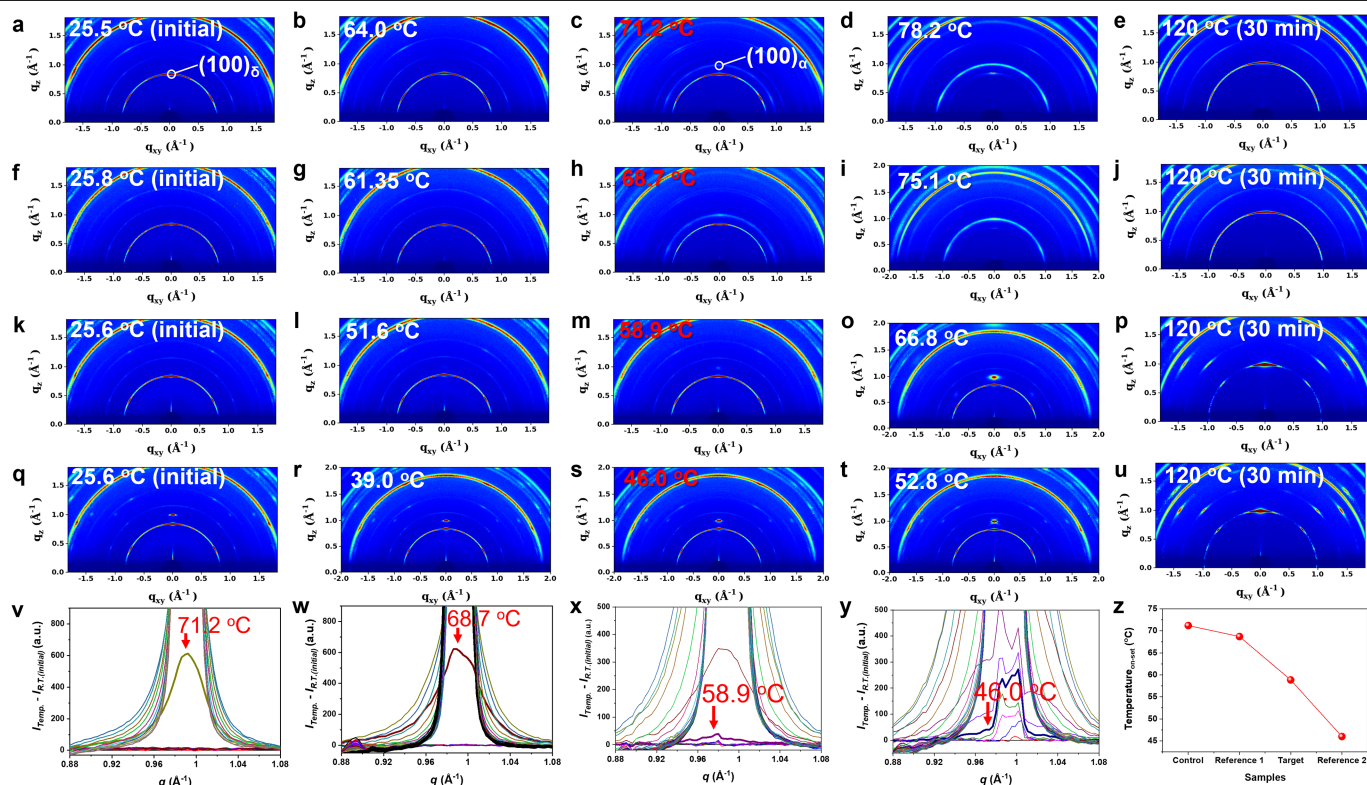
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Extended Data Fig. 1 | Surface morphologies of thin films. Thin films spin-coated with 10 mol% MACl (a) and BACl (b) added to FAPbI₃ precursor containing 35 mol% MACl. Scale bars, 1 μ m.

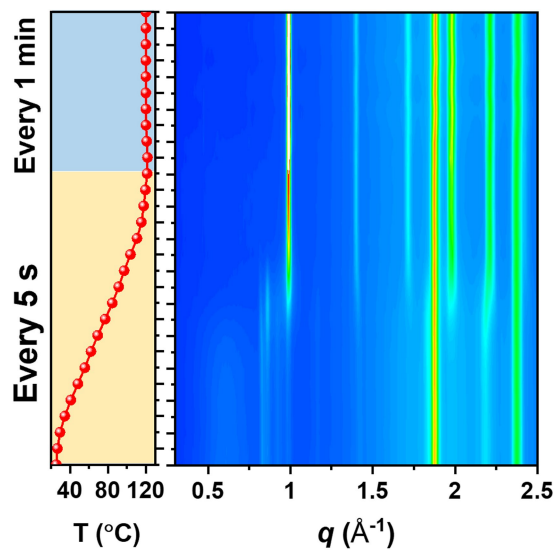


Extended Data Fig. 2 | Changes in the width of the (100)_a peak with time. Shown during annealing from RT to 120 °C, using the thin films formed by dripping the antisolvent during spin-coating of the FAPbI₃ precursor solutions containing 0, 5, 10 or 15 mol% PAcl added to a 35 mol% MACl.

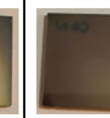
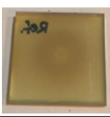
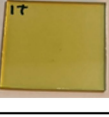
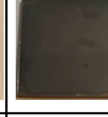
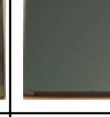
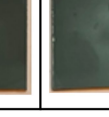


Extended Data Fig. 3 | Representative 2D GI-WAXD images and enlarged 1D GI-WAXD profiles during in situ experiments. a–e, Control. f–j, Target. k–p, Reference 1. q–u, Reference 2. GI-WAXD measured while increasing and maintaining the temperature from RT to 120 °C. v–y, Enlarged 1D GI-WAXD profiles of control (v), reference 1 (w), target (x) and reference 2 (y) samples around the $(100)_\alpha$ peak to investigate the onset temperature of the δ -phase to α -phase transition. z, Plot of the onset temperature of the δ -phase to α -phase transition. 2D GI-WAXD shows that the transition temperature from δ to α phase gradually decreases as the length of the alkyl group in RACI increases

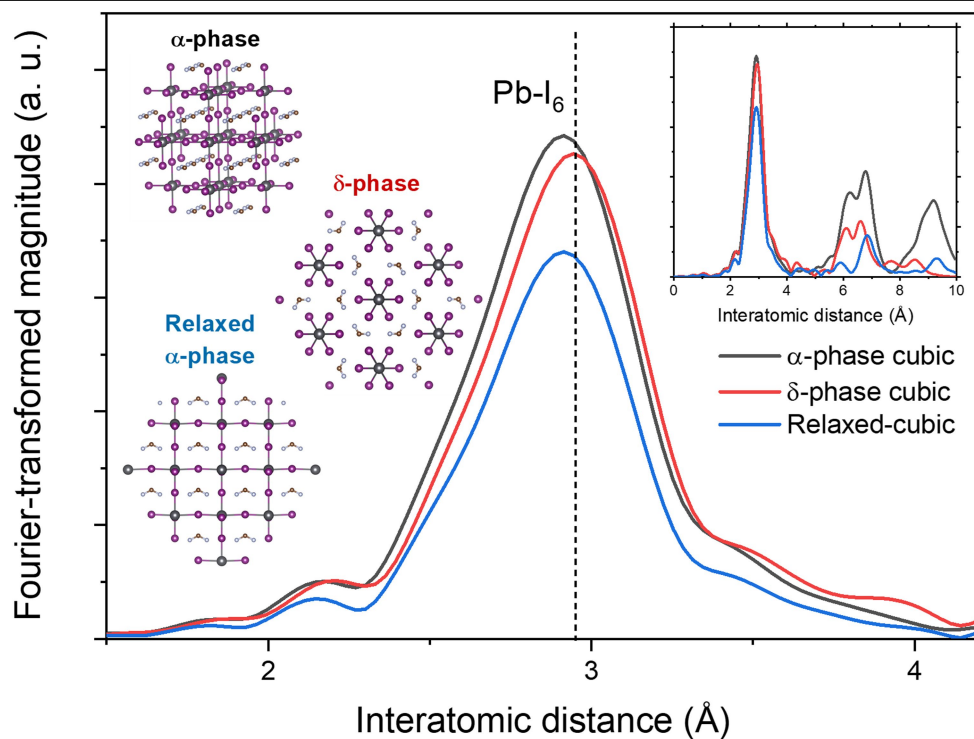
from MACI to PACI and BACI. The boiling points and basicity of deprotonated MA^0 , PA^0 and BA^0 are different. Thus, strong acid–base interaction can occur between the Lewis acid PbI_2 and the Lewis base R-NH_2 on the surface of FAPbI_3 . Alkyl substituents such as methyl, propyl and butyl are electron-donating groups and tend to increase with increasing length. This makes BA^0 more Lewis basic, increasing $\text{RH}_2\text{N-PbI}_2$ interaction. Eventually, BA^0 deprotonated in BACI lowers more surface energy of δ - FAPbI_3 , leading to a faster and more effective δ -phase to α -phase transition.



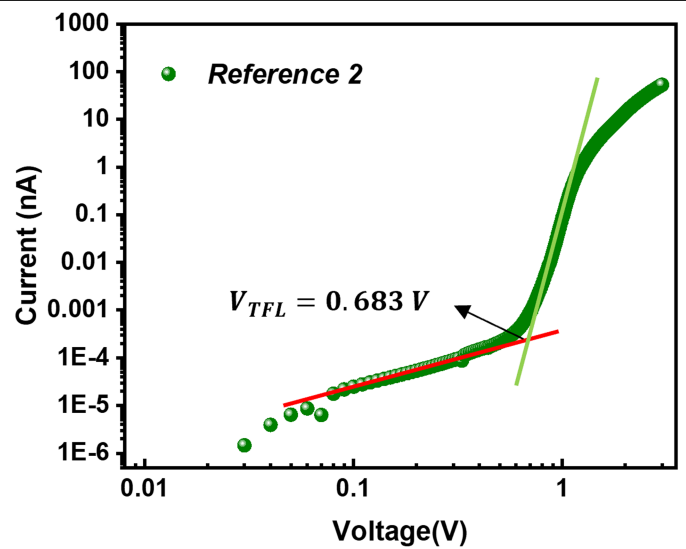
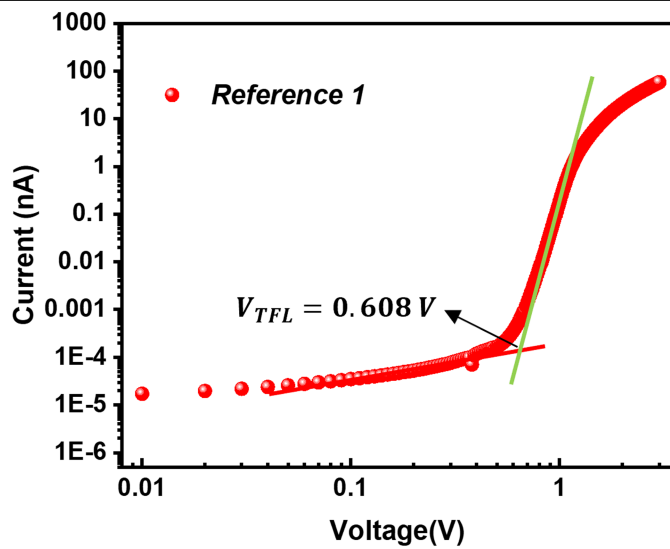
Extended Data Fig. 4 | In situ GI-WAXD. Shown during heating from RT to 120 °C using the thin film deposited without dripping the antisolvent during spin-coating of the FAPbI₃ precursor solutions with 10 mol% PAcI added to a 35 mol% MACI.

Time (min) mol%	0	10	20	30	40	50	60	70	120
Control									
Reference 1									
Target									
Reference 2									

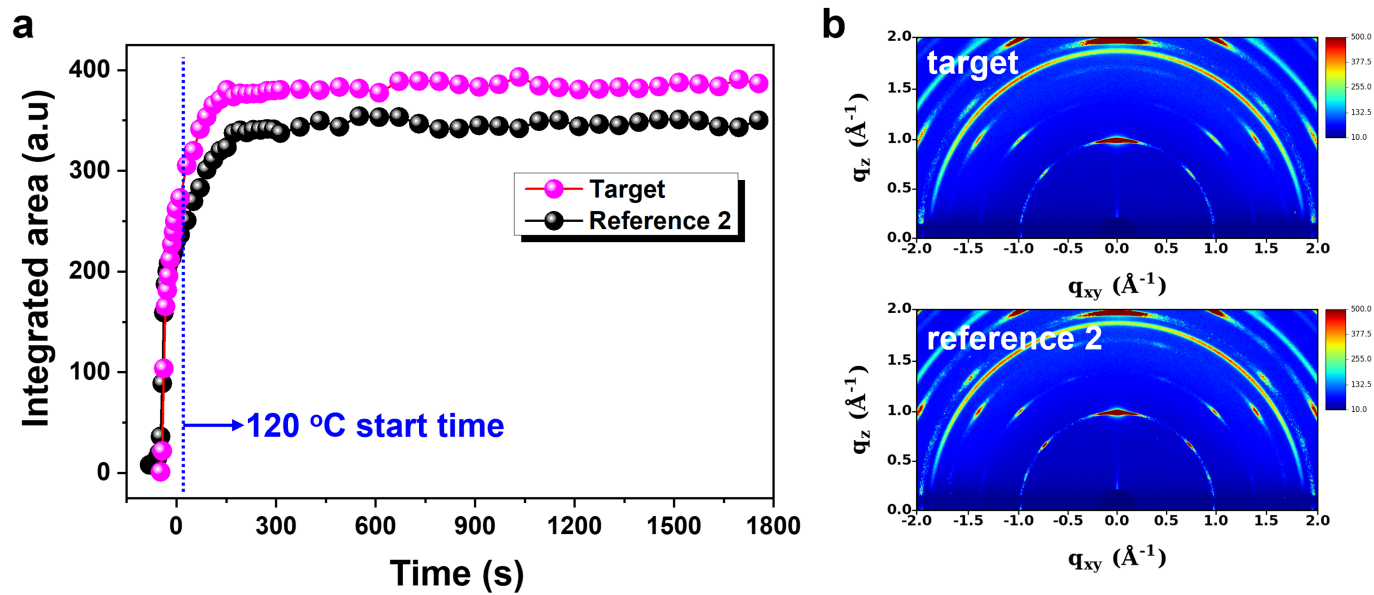
Extended Data Fig. 5 | Changes in colour observed at RT of thin films. Thin films were deposited by dripping the antisolvent during spin-coating of the FAPbI₃ precursor solutions containing 10 mol% of MACl, PAcI and BAcI, respectively, added to a 35 mol% of MACl.



Extended Data Fig. 6 | Fourier-transformed RDF. RDF obtained using theoretically calculated Pb L_{III} -edge EXAFS for the α -FAPbI₃, δ -FAPbI₃ and relaxed-FAPbI₃, respectively.



Extended Data Fig. 7 | Space-charge-limited current analysis. Shown for reference 1 and reference 2. V_{TFL} is trap-filled limit voltage.



Extended Data Fig. 8 | Changes in the mutually normalized integrated area over time and representative 2D GI-WAXD images. a. Changes in the mutually normalized integrated area obtained by α -phase peaks of in situ

GI-WAXD measured while increasing and maintaining the temperature from RT to 120 °C. **b.** Representative 2D GI-WAXD images of target and reference 2 samples annealed at 120 °C for 30 min.

Solar Cells Reporting Summary

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	Experimental Methods
Method used to determine the device area	☒ Yes ☐ No	Experimental Methods

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	Experimental Methods and Figure 4b
Voltage scan conditions <i>For instance: scan direction, speed, dwell times</i>	☒ Yes ☐ No	Experimental Methods
Test environment <i>For instance: characterization temperature, in air or in glove box</i>	☒ Yes ☐ No	Experimental Methods and Main Text
Protocol for preconditioning of the device before its characterization	☒ Yes ☐ No	Experimental Methods
Stability of the J-V characteristic <i>Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.</i>	☒ Yes ☐ No	Experimental Methods and Main Text

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes ☒ No	No unusual hysteric behavior is observed
Related experimental data	☐ Yes ☒ No	No unusual hysteric behavior is observed

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	Figure 4c
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	Figure 4c
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	<i>Explain why this information is not reported/not relevant.</i>

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Experimental Methods
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Experimental Methods and Main Text

Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	Certified values were displayed
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	Experimental Methods
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	The mask size was not varied
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	Supplementary Figure 12
A copy of any certificate(s) <i>Provide in Supplementary Information</i>	☒ Yes ☐ No	Supplementary Figure 12
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Experimental Methods
Statistical analysis of the device performance	☒ Yes ☐ No	Figure 4a
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions <i>For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature</i>	☒ Yes ☐ No	Figure 4d